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(54) **VOCALIZATION ASSISTANCE DEVICE**

(71) Applicant: **National University Corporation Tokyo Medical and Dental University**, Tokyo (JP)

(72) Inventors: **Haruka Tohara**, Tokyo (JP); **Taishi Yamada**, Tokyo (JP); **Kazuharu Nakagawa**, Tokyo (JP); **Kohei Yamaguchi**, Tokyo (JP); **Daiki Mizuguchi**, Tokyo (JP)

(73) Assignee: **National University Corporation Tokyo Medical and Dental University**, Tokyo (JP)

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H04R 1/02 (2006.01)

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(58) **Field of Classification Search**
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USPC 381/70, 334
See application file for complete search history.

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Primary Examiner — Ammar T Hamid

(57) **ABSTRACT**

A vocalization assistance device includes: a mounting body that is mounted in an oral cavity of a person and that has a facing portion facing a palate of the person; a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound; and a passage that is provided at the facing portion and that is configured to transmit the original sound output from the diaphragm to a rear side of the oral cavity.

14 Claims, 10 Drawing Sheets

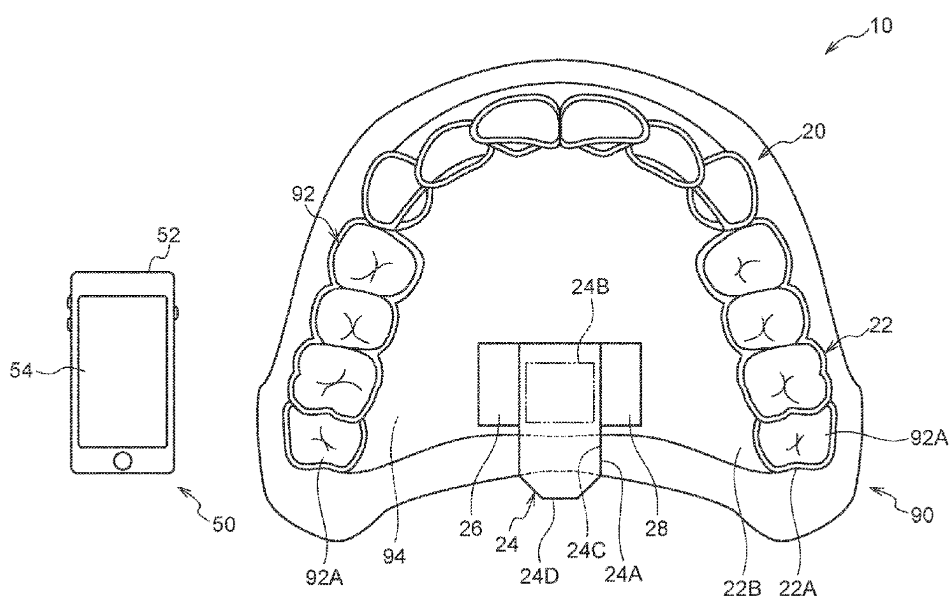


FIG.1

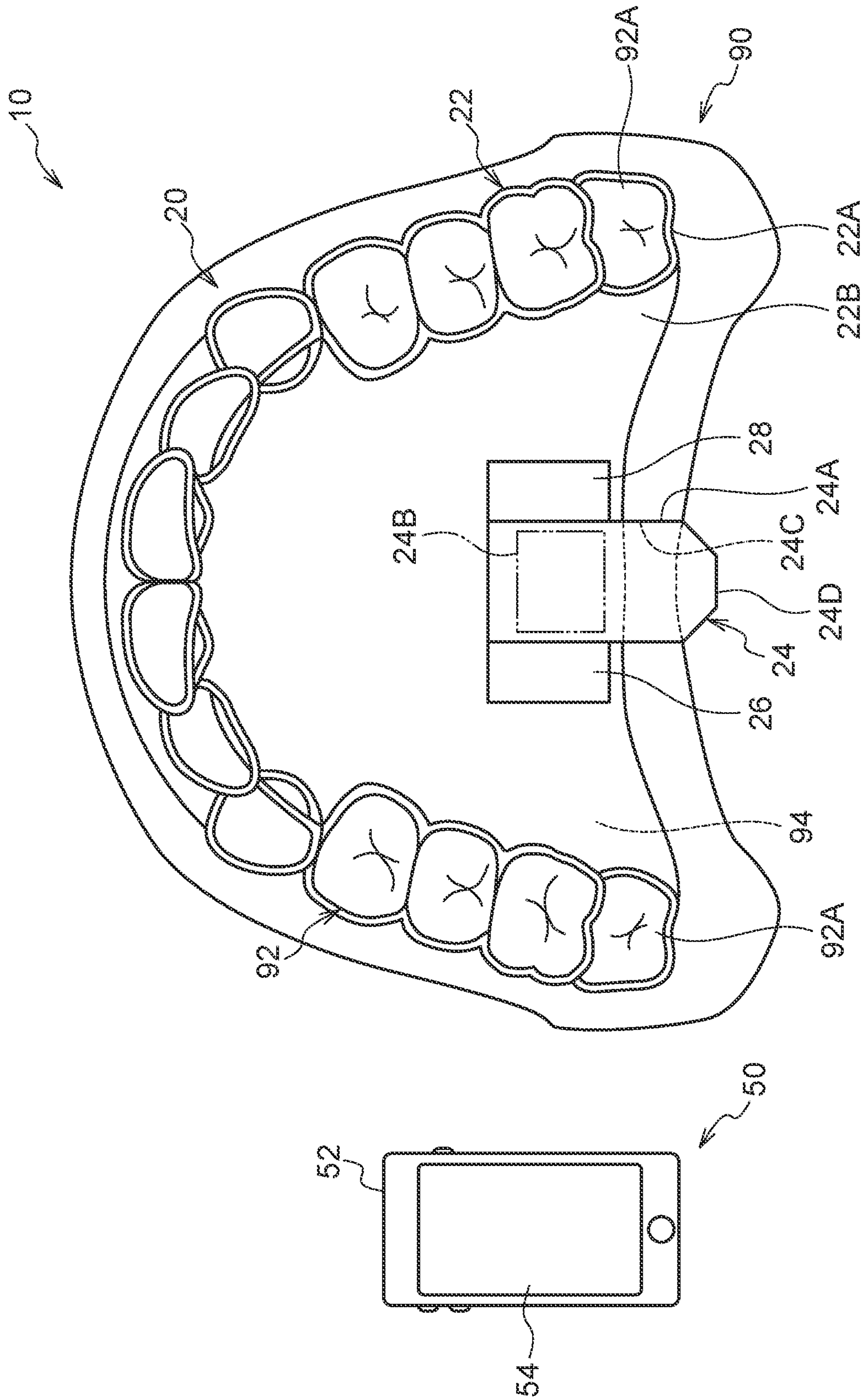


FIG.2

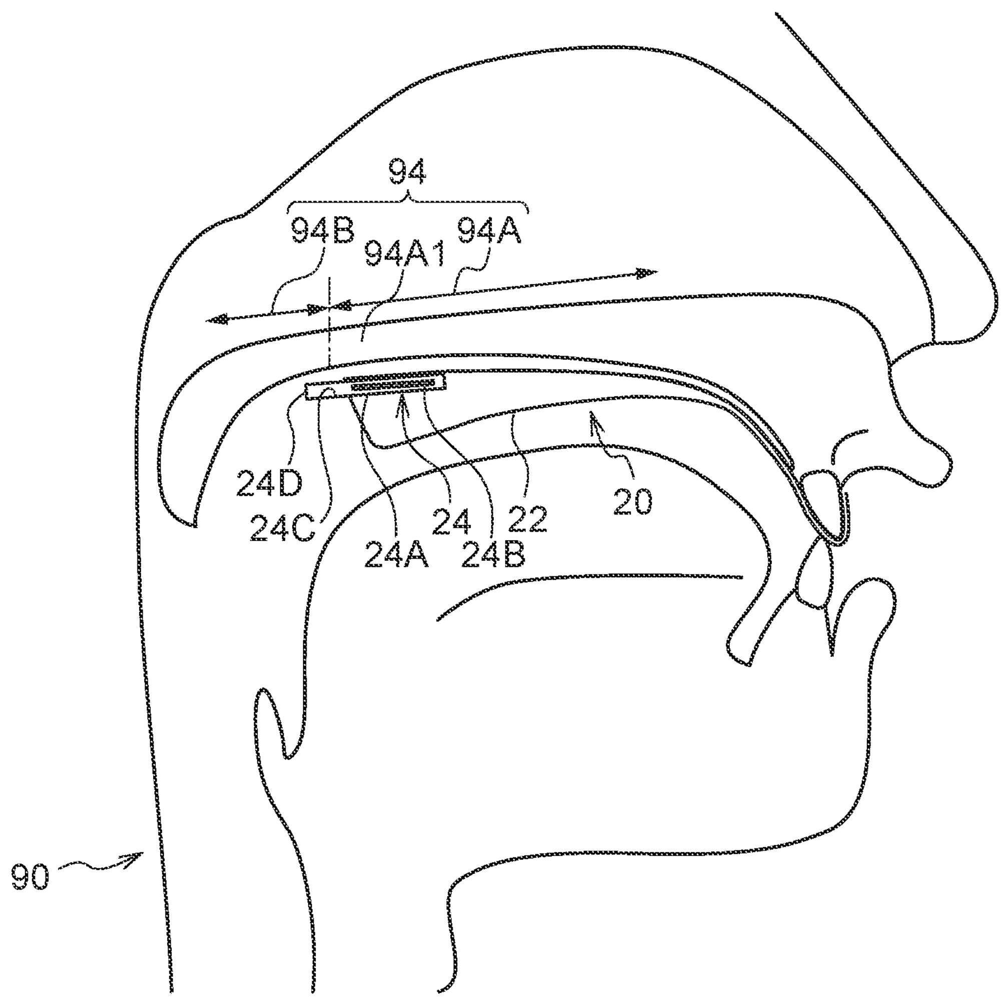
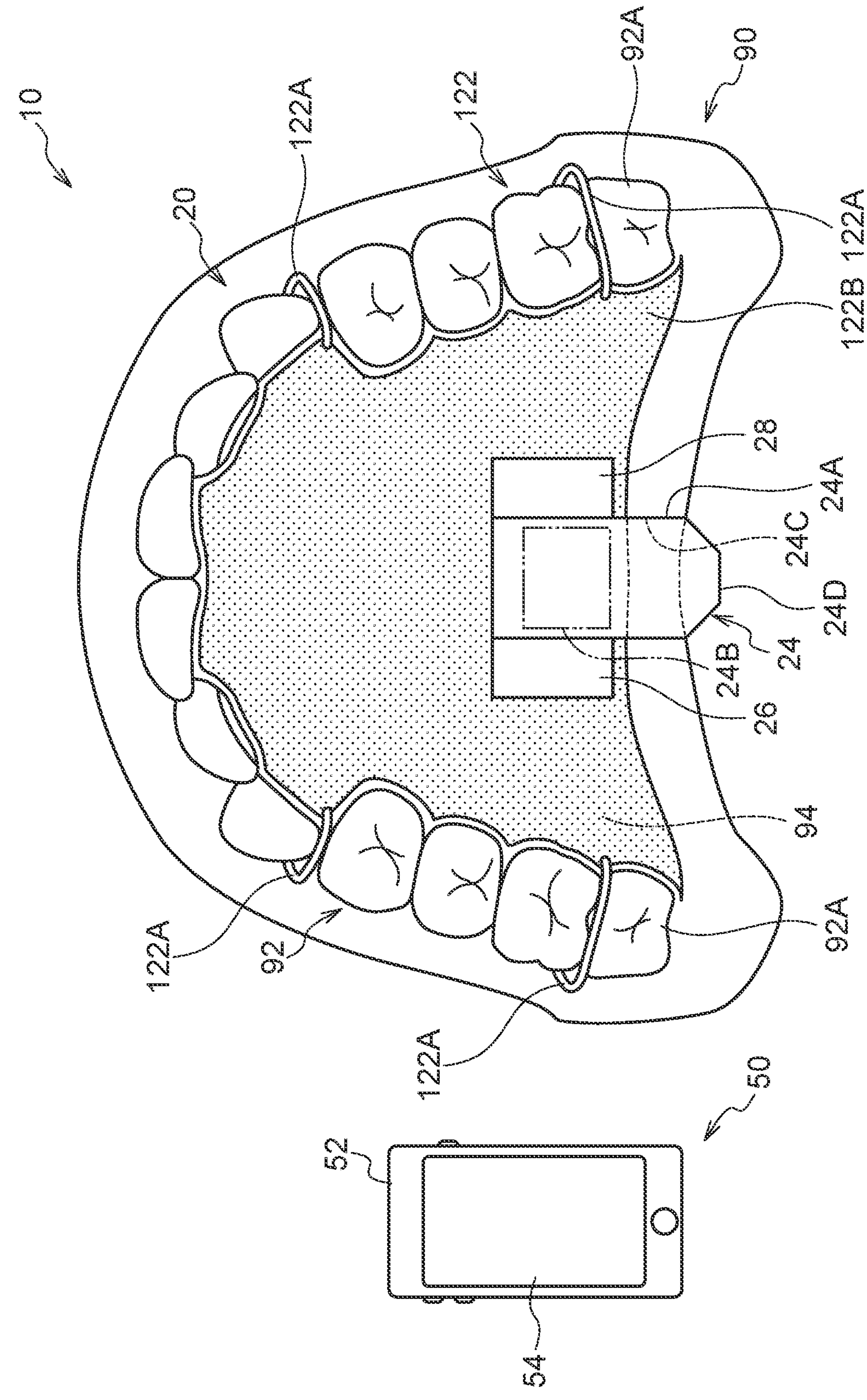


FIG.3


10
20
122A
122
92A
90
122A
122B
28
24A
24C
24D
24
24B
26
94
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122A
92
122A
50
52
54

FIG.4

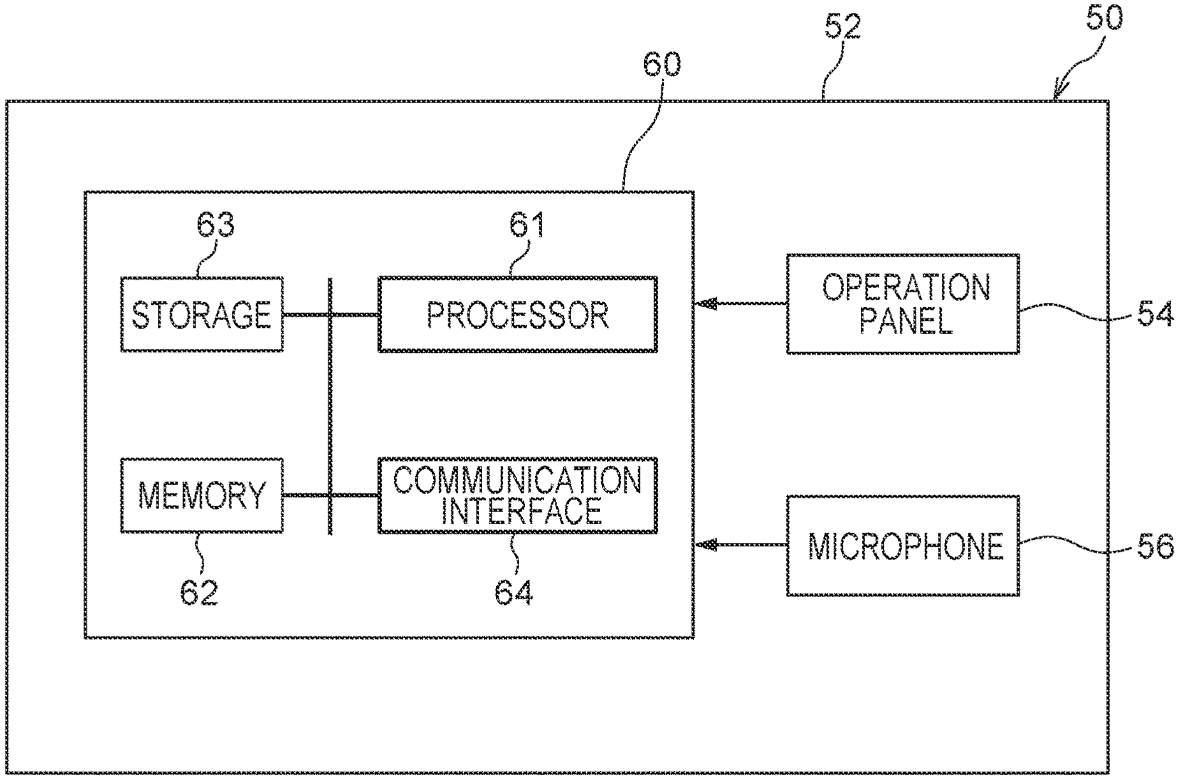

52
50
60
63
61
STORAGE
PROCESSOR
OPERATION PANEL
54
MEMORY
COMMUNICATION INTERFACE
MICROPHONE
56
62
64

FIG.5

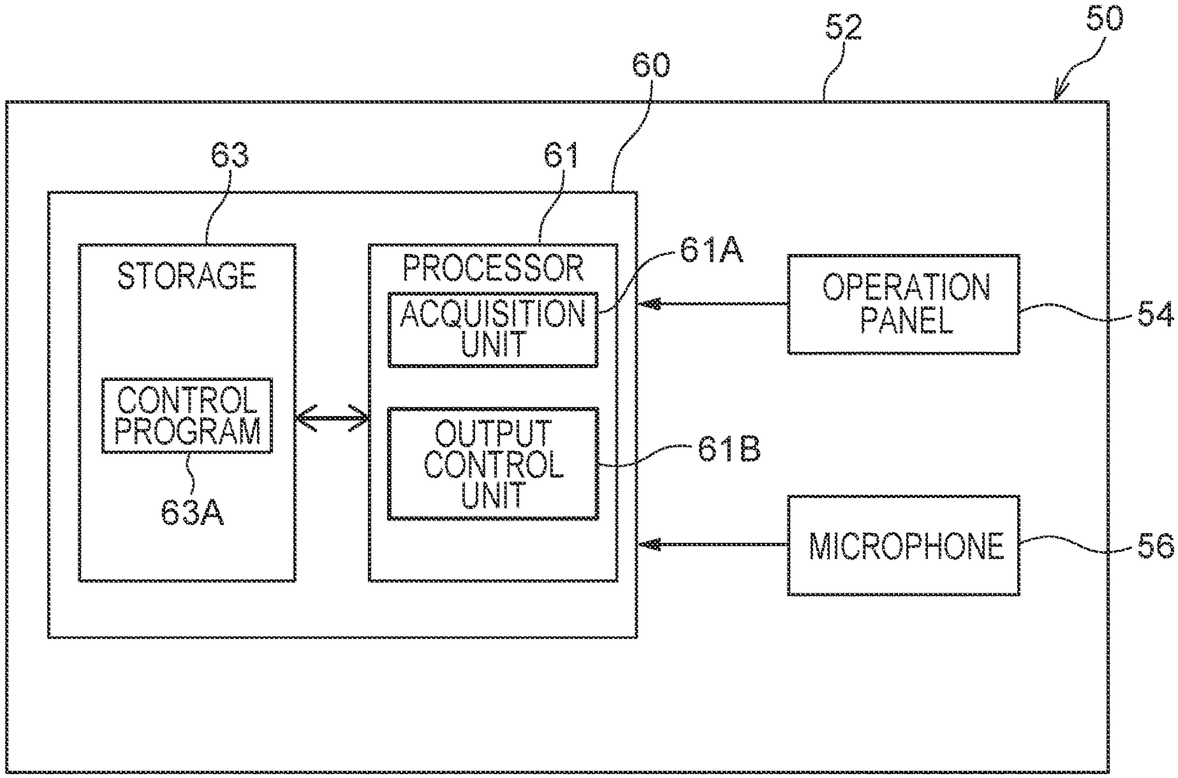

50
52
60
63
61
STORAGE
PROCESSOR
61A
ACQUISITION UNIT
OPERATION PANEL
54
CONTROL PROGRAM
OUTPUT CONTROL UNIT
61B
63A
MICROPHONE
56

FIG.6

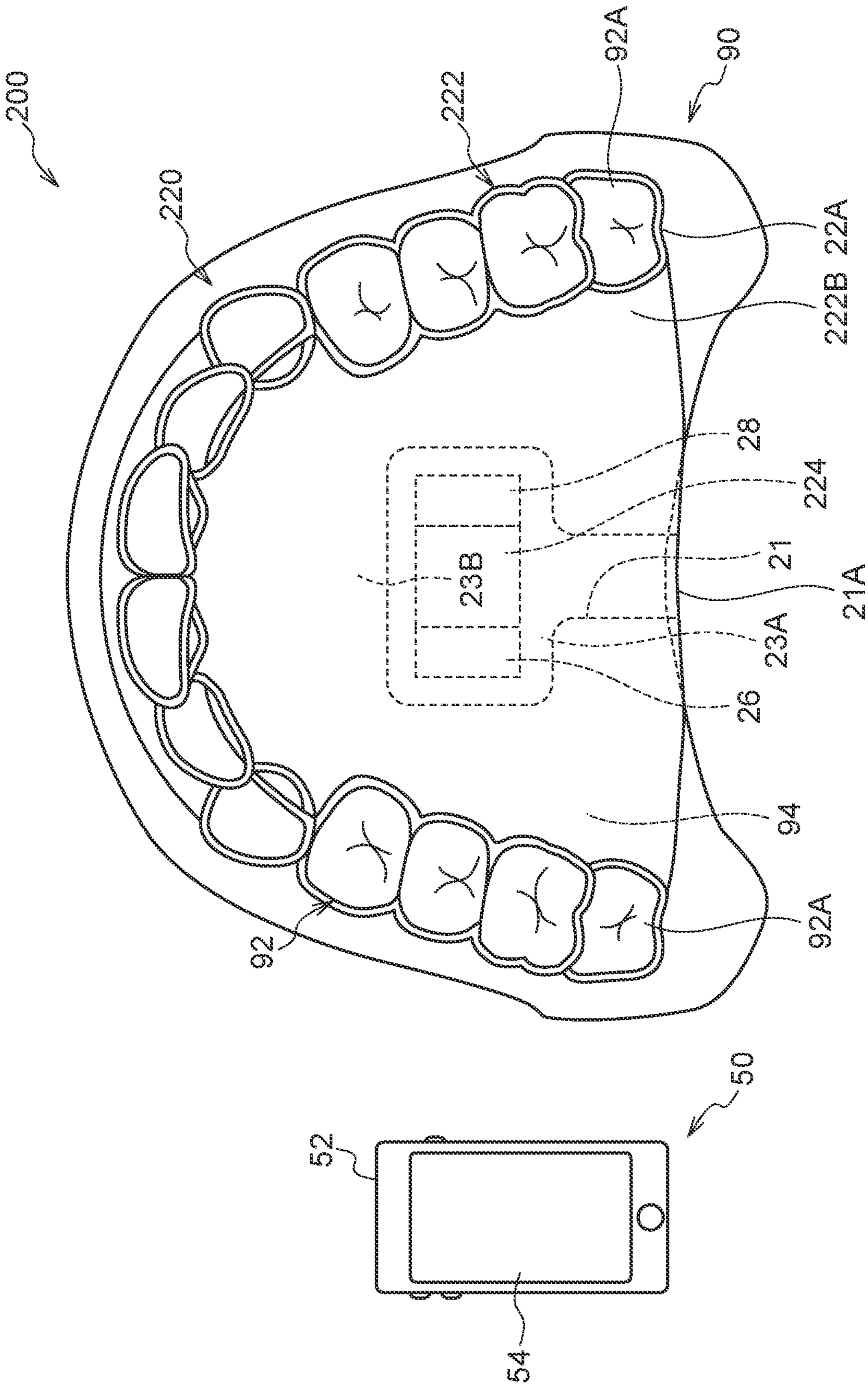
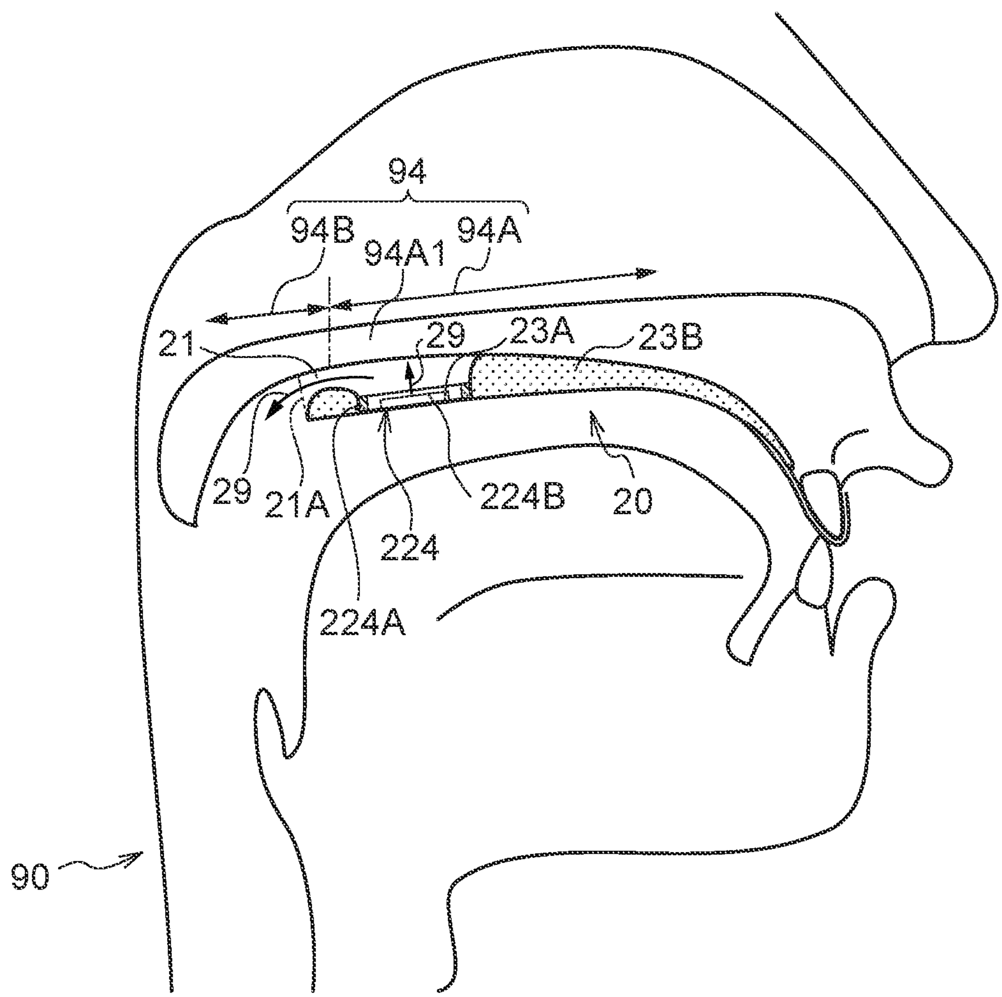

200
220
222
92A
90
22A
222B
28
224
23B
21
21A
23A
26
94
92
92A
50
52
54

FIG.7



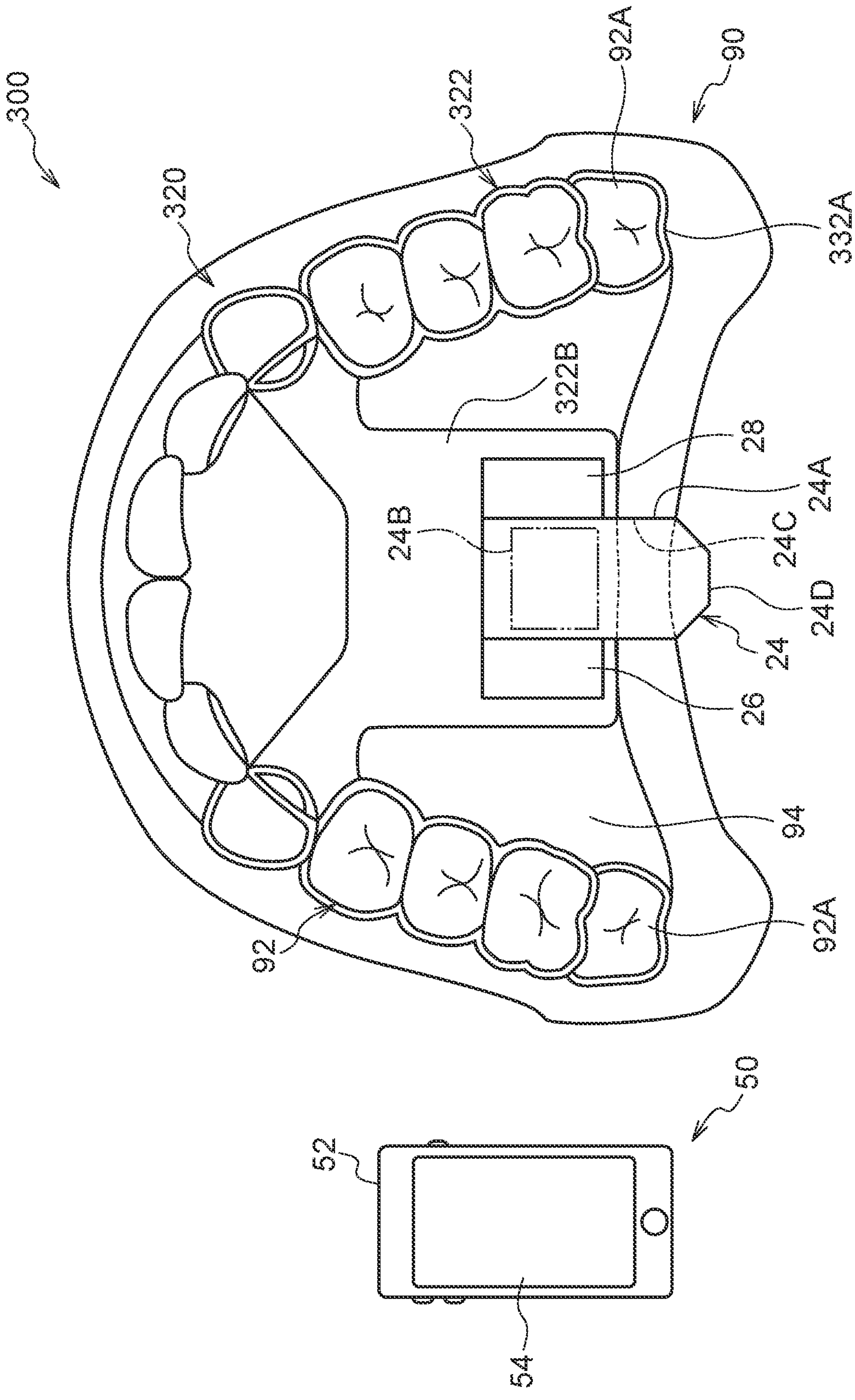

300
320
322
92A
90
332A
322B
28
24A
24B
24C
24D
24
26
94
92A
92
50
52
54

FIG.8

FIG.9

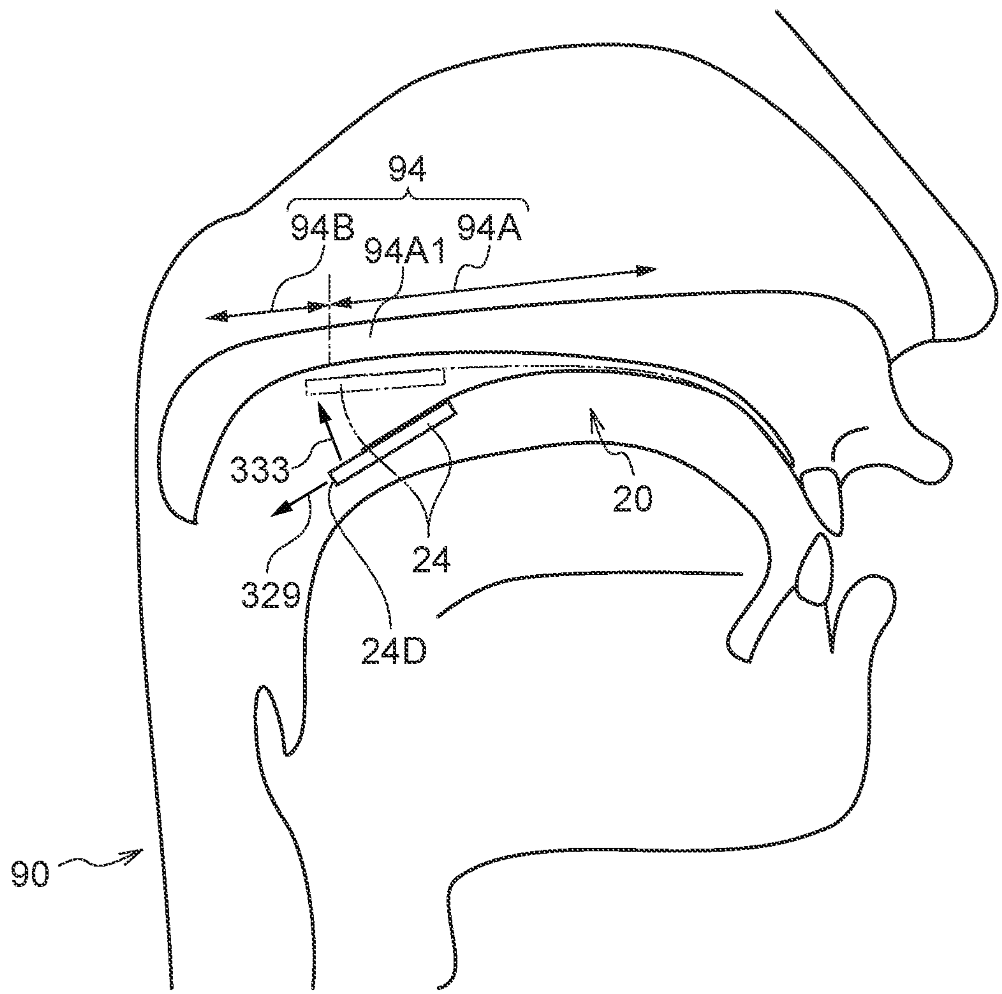
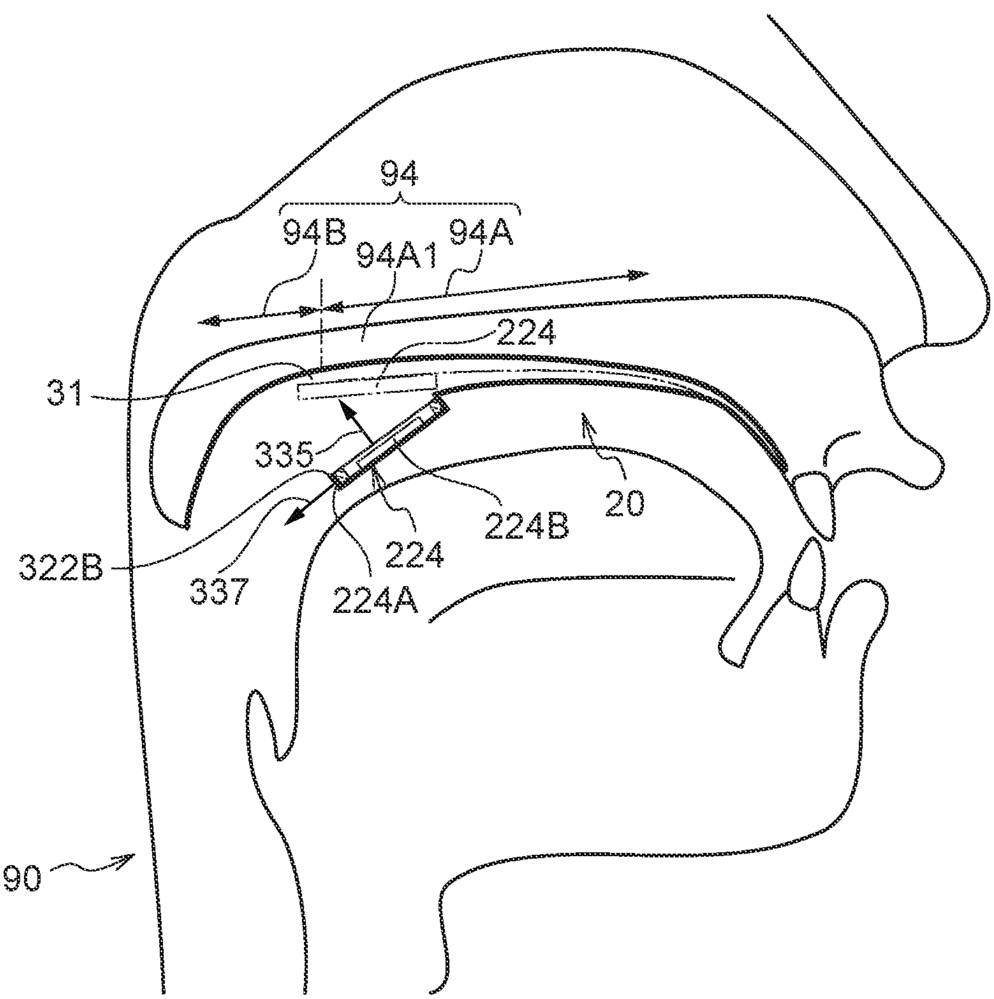


FIG.10



VOCALIZATION ASSISTANCE DEVICE

RELATED APPLICATIONS

This application is a National Phase of PCT Patent Application No. PCT/JP2022/002720 having International filing date of Jan. 25, 2022, which claims the benefit of priority of Japan Patent Application No. 2021-010291 filed on Jan. 26, 2021. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION

The present disclosure relates to a vocalization assistance device.

Examples of a vocalization assistance device include vocalization assistance devices such as artificial larynx disclosed in JP8-501950A, JP2015-192851A, JP8-24688A, and KR10-2012-0034395A.

The artificial larynx disclosed in JP8-501950A includes a first unit attached into a mouth, having a denture, a loudspeaker, a power amplifier, a built-in power source, and a wireless receiver, an input control device, a built-in power source, and a handheld second unit having an electronic circuit and a transmitter that allow a user to modify the frequency and volume generated by the units in the mouth.

The vocalization assistance device disclosed in JP2015-192851A includes a speaker, a housing that transmits a sound generated by the speaker to a nostril of a person, a sound generator that causes the speaker to generate a sound including a frequency component that causes resonance in an oral cavity of the person, and a switch that switches whether or not the sound generator causes the speaker to generate a sound.

The electric artificial larynx disclosed in JP8-24688A includes a biological information detection unit that detects biological information, and a control unit that controls a basic frequency and a volume of a substitute sound source output from an acoustic transducer according to the biological information detected by the biological information detection unit.

KR10-2012-0034395A discloses an artificial larynx device in which a speaker is disposed at the center of the palate.

SUMMARY OF INVENTION

For example, in the case of making an explosive sound made by releasing the tongue brought into contact with the palate, if the original sound that is the source of the voice is produced on the front side of the position where the tongue comes into contact with and separates from the palate, the original sound cannot be blocked, and thus the utterance of the explosive sound is difficult. Therefore, the original sound desirably is produced on the rear side of the oral cavity.

In view of the above fact, an object of the present disclosure is to provide a vocalization assistance device capable of producing an original sound on a rear side in an oral cavity.

A vocalization assistance device of the present disclosure includes: a mounting body that is mounted in an oral cavity of a person and that has a facing portion facing a palate of the person; a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound; and a passage that is provided at the facing portion and that is configured to transmit the original sound output from the diaphragm to a rear side of the oral cavity.

A vocalization assistance device of the present disclosure includes: a mounting body that is mounted in an oral cavity of a person, that has a facing portion facing a palate of the person, and that is movable in a vertical direction with a front side as a fulcrum; and a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound.

Advantageous Effects of Invention

As described above, the vocalization assistance device of the present disclosure has an excellent effect that the original sound can be produced on the rear side of the oral cavity.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. **1** is a schematic diagram illustrating an example of a configuration of a vocalization assistance device according to a first embodiment.

FIG. **2** is a side view illustrating an example of a configuration of the vocalization assistance device according to the first embodiment.

FIG. **3** is a schematic view illustrating an example of a modification of the mounting body in the vocalization assistance device according to the first embodiment.

FIG. **4** is a block diagram illustrating an example of a hardware configuration of an external device according to the first embodiment.

FIG. **5** is a block diagram illustrating an example of a functional configuration of a processor of the external device according to the first embodiment.

FIG. **6** is a schematic diagram illustrating an example of a configuration of a vocalization assistance device according to a second embodiment.

FIG. **7** is a side view illustrating an example of a configuration of the vocalization assistance device according to the second embodiment.

FIG. **8** is a schematic diagram illustrating an example of a configuration of a vocalization assistance device according to a third embodiment.

FIG. **9** is a side view illustrating an example of a configuration of the vocalization assistance device according to the third embodiment.

FIG. **10** is a side view illustrating an example of a modification of a speaker in the vocalization assistance device according to the third embodiment.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

Hereinafter, an example of an embodiment according to the present disclosure will be described with reference to the drawings.

First Embodiment

A configuration of a vocalization assistance device **10** according to a first embodiment will be described. FIG. **1** is a schematic view of the vocalization assistance device **10** according to the present embodiment.

The vocalization assistance device **10** illustrated in FIG. **1** is a device that assists a person's vocalization. Specifically, the vocalization assistance device **10** generates an original sound in the oral cavity of a person. As a result, the original sound generated in the oral cavity is articulated into a verbal sound (that is, a consonant and a vowel) by an articulatory organ such as a lip and a tongue, whereby utterance is performed. In the present specification, the "original sound" refers to a sound that is a source of voice.

As an example, the vocalization assistance device **10** is used by a person who has extracted the larynx due to illness or the like, a person who has incised the trachea due to the use of a ventilator, or the like. Note that the vocalization assistance device **10** may be used as a device (that is, a voice changer) that can change the voice and speak, a musical instrument similar to a talking modulator, and the like. Therefore, the vocalization assistance device **10** may be used by a healthy person.

In the present embodiment, as illustrated in FIG. **1**, the vocalization assistance device **10** includes an internal device **20** and an external device **50**. Hereinafter, specific configurations of the internal device **20** and the external device **50** will be described.

<Internal Device **20**>

The internal device **20** is a device mounted in the oral cavity of a person. Therefore, the internal device **20** can also be said to be a mounting device. Hereinafter, the person wearing the internal device **20** may be referred to as a wearer **90**. As illustrated in FIG. **1**, the internal device **20** includes a mouthpiece **22**, a speaker **24**, a power source **26**, and a communication unit **28**.

<Mouthpiece **22**>

The mouthpiece **22** is a member mounted in the oral cavity of the wearer **90**, and is an example of the mounting body. The mouthpiece **22** is mounted in the oral cavity of the wearer **90** in a predetermined direction. More specifically, the mouthpiece **22** is mounted at a predetermined position in the oral cavity of the wearer **90**. Specifically, the mouthpiece **22** includes a mounting portion **22**A attached to an upper dentition **92** of the wearer **90**, and an attachment portion **22**B facing the palate **94**. The attachment portion **22**B is an example of a facing portion.

The mounting portion **22**A is formed in a U shape in which a rear side of an oral cavity of the wearer **90** is open in a bottom view. Note that the bottom view in the present specification refers to a case of being viewed from the lower side to the upper side in a state where the mouthpiece **22** is mounted in the oral cavity of the wearer **90** (hereinafter, referred to as a "mounting state of the mouthpiece **22**"). The mounting portion **22**A is mounted on the dentition **92** so as to cover the outer peripheral surface, the inner peripheral surface, and the occlusal surface of the dentition **92**. In the following description, the up-down direction, the left-right direction, and the front-rear direction are directions in the oral cavity in the mounting state of the mouthpiece **22**.

The attachment portion **22**B is disposed along the palate **94** and is connected to the upper end of the mounting portion **22**A on the inner peripheral side of the mounting portion **22**A. The attachment portion **22**B is a portion to which components such as the speaker **24** are attached. It can also be said that the mounting portion **22**A has a function of holding a state where the attachment portion **22**B is disposed on the palate **94**. The attachment portion **22**B may be disposed so as to face the palate **94**, and is not necessarily in contact with the palate **94**, and may have a gap with respect to the palate **94**.

As a material of the mouthpiece **22**, for example, a resin material such as ethylene vinyl acetate (EVA), polyurethane, or polyethylene (PE) is used. A material of the mouthpiece **22** is not limited to the above-mentioned materials, and various materials can be used. The mouthpiece **22** may be a mouthpiece adapted to a tooth shape of the wearer **90**, or may be a general-purpose mouthpiece having a standard shape.

As an example of the mounting body, the attachment portion **22**B may be disposed so as to face the palate **94**. Therefore, the mounting portion **22**A may not be mounted on all the teeth of the dentition **92**, and may be mounted on some teeth (for example, a plurality of teeth on the rear side) of the dentition **92**.

<Speaker **24**>

The speaker **24** is a device that converts an electrical signal (that is, electrical vibration) into a sound (that is, physical vibration). The speaker **24** includes a case **24**A, a diaphragm **24**B, a passage **24**C, and an amplifier (not illustrated). The speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B of the mouthpiece **22**. Thus, each part (specifically, the diaphragm **24**B, the passage **24**C, and the amplifier (not illustrated)) of the speaker **24** is provided at the attachment portion **22**B.

The case **24**A is formed in a flat shape that is thin in the vertical direction and expands in the horizontal direction and the front-rear direction. That is, the case **24**A is formed in a plate shape whose thickness direction is the vertical direction.

The diaphragm **24**B is accommodated in the case **24**A. The diaphragm **24**B converts an electrical signal into a sound and outputs an original sound. An outlet **24**D through which the original sound is emitted is formed at a rear end of the case **24**A.

The passage **24**C is formed inside the case **24**A. Specifically, the passage **24**C is formed by being surrounded in the vertical and horizontal directions in the oral cavity by the case **24**A. That is, the case **24**A has an upper wall, a lower wall, a left wall, and a right wall forming the passage **24**C. The passage **24**C extends rearward in the oral cavity from the diaphragm **24**B. The passage **24**C transmits the original sound output from the diaphragm **24**B to the rear side of the oral cavity. The passage **24**C has the outlet **24**D formed at the rear end of the case **24**A. The passage **24**C emits the original sound transmitted to the rear side of the oral cavity from the outlet **24**D.

In the speaker **24**, the electrical signal received by the communication unit **28** is amplified by an amplifier (not illustrated), the diaphragm **24**B converts the electrical signal into the original sound, and the passage **24**C emits the original sound to the rear side of the oral cavity through the outlet **24**D.

The size of the speaker **24** (specifically, the case **24**A) in a bottom view is at least smaller than the size of the attachment portion **22**B in a bottom view. Specifically, the speaker **24** (specifically, the case **24**A) has at least a width in the left-right direction smaller than a width in the left-right direction of the attachment portion **22**B. In other words, the speaker **24** (specifically, the case **24**A) has at least a width in the left-right direction narrower than an interval in the left-right direction between the left and right rearmost teeth **92**A of the wearer **90**. The rearmost tooth **92**A is a tooth disposed on the rearmost side in the dentition **92**. In the example of the present embodiment, the rearmost tooth **92**A is a second molar tooth. In the present embodiment, the length of the speaker **24** in the front-rear direction is shorter than the length of the attachment portion **22**B in the front-rear direction.

The speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B such that the outlet **24**D is disposed on the rear side in the oral cavity of the wearer **90** in the mounting state of the mouthpiece **22** (see FIG. **2**).

In the present embodiment, the outlet **24**D is disposed at a position on the rear side from a rear end **94A1** of a hard palate **94**A of the wearer **90**. In other words, the speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B such that, in the mounting state of the mouthpiece **22**, the outlet **24**D is disposed at a position on the rear side of the rear end **94A1** of the hard palate **94**A of the wearer **90** (see FIG. **2**). The position on the rear side of the rear end **94A1** of the hard palate **94**A is a position including the rear end **94A1** of the hard palate **94**A and a position on the rear side of the rear end **94A1** of the hard palate **94**A (that is, a soft palate **94**B present on the rear side of the hard palate **94**A). In the present embodiment, specifically, the outlet **24**D is disposed at a position on the rear side of the rear end **94A1** of the hard palate **94**A.

More specifically, the outlet **24**D is disposed at a position on the rear side from rearmost tooth **92**A of the wearer **90**. In other words, the speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B such that, in the mounting state of the mouthpiece **22**, the outlet **24**D is disposed at a position on the rear side from the rearmost tooth **92**A of the wearer **90** (see FIG. **1**). The position on the rear side from the rearmost tooth **92**A is a position including the rearmost tooth **92**A and the rear side of the rearmost tooth **92**A. For the wearer **90** having no second molar tooth, in the case where the second molar tooth is present, the speaker **24** is attached to the attachment portion **22**B such that the outlet **24**D is disposed at a position on the rear side of the position where the second molar tooth is disposed.

Furthermore, the outlet **24**D is disposed at a position on the rear side from the contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of consonant [k]. In other words, the speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B such that, in the mounting state of the mouthpiece **22**, the outlet **24**D is disposed at a position on the rear side from a contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of consonant [k]. The position on the rear side from the contact position is a position including the contact position and a position on the rear side of the contact position.

Furthermore, in the present embodiment, the outlet **24**D is disposed at a position on the rear side from the rear end of the attachment portion **22**B. In other words, the speaker **24** (specifically, the case **24**A) is attached to the attachment portion **22**B such that, in the mounting state of the mouthpiece **22**, the outlet **24**D is disposed at a position on the rear side from the rear end of the attachment portion **22**B (see FIG. **1**). The position on the rear side from the rear end of the attachment portion **22**B is a position including the rear end of the attachment portion **22**B and a position on the rear side of the rear end of the attachment portion **22**B.

Specifically, in the present embodiment, the case **24**A protrudes rearward from the attachment portion **22**B. Therefore, in the present embodiment, the outlet **24**D formed in the case **24**A is disposed at a position on the rear side of the rear end of the attachment portion **22**B.

The rearward protruding portion of the case **24**A is not in contact with the palate **94**. That is, the case **24**A is attached to the attachment portion **22**B such that the protruding portion is not in contact with the palate **94**.

The entire case **24**A including a protruding portion protruding rearward from the attachment portion **22**B is not in contact with the palate **94**. The speaker **24** is desirably disposed on the rear side in the oral cavity of the wearer **90** as far as possible within a range in which the pharyngeal reflex does not occur in the mounting state of the mouthpiece **22**. Incidentally, the pharyngeal reflex is a reflex response that occurs when various sites in the oral cavity are stimulated, and is a reflex response that feels sickening (that is, causes nausea).

Note that the position of the hard palate **94**A, the position of the rearmost tooth **92**A, the position where the tongue comes into contact with the palate **94** when making the sound of consonant [k], the position where the pharyngeal reflex occurs, and the like vary from person to person. Therefore, for example, the position of the outlet **24**D or the like is adjusted in accordance with the individual wearer **90**, and the speaker **24** (specifically, the case **24**A) is attached to the mouthpiece **22**.

A plurality of speakers **24** may be provided at the attachment portion **22**B. As a result, the range of sound that can be made can be expanded. Furthermore, the speaker **24** may include a plurality of diaphragms **24**B. As a result, the range of sound that can be made can be expanded.

<Power Source **26**>

The power source **26** is a power source for driving the speaker **24** (specifically, the above-described amplifier or the like). As the power source **26**, a small rechargeable battery or the like is used. The power source **26** is attached to the attachment portion **22**B together with the speaker **24**. The power source **26** is disposed at a position close to (for example, adjacent to) the speaker **24**.

<Communication Unit **28**>

The communication unit **28** is a component for communicating with the external device **50**. In the present embodiment, the communication unit **28** receives an electrical signal converted into sound by the speaker **24** by wireless communication means. As an example of a wireless communication means, a wireless communication function such as Bluetooth (registered trademark) is used. The communication unit **28** is attached to the attachment portion **22**B together with the power source **26** and the speaker **24**. The communication unit **28** is disposed at a position close to (for example, adjacent to) the speaker **24**.

Note that the amplifiers in the communication unit **28**, the power source **26**, and the speaker **24** may be disposed outside the oral cavity. For example, components of the communication unit **28**, the power source **26**, and the amplifier are accommodated in a case. The case can be configured as, for example, an ear hook type that can be hung on the ear. In this case, the wiring that connects the components accommodated in the case and the speaker **24** in the oral cavity is disposed, for example, into the oral cavity through the corners of the mouth while extending from the case through the ears and the cheeks. Furthermore, inside the oral cavity, the wiring disposed into the oral cavity through the corners of the mouth is disposed, for example, such that the wiring on the outer side of the dentition **92** is disposed on the rear side, is folded back in a U shape at the rearmost tooth **92**A, and is connected to the speaker **24**.

Furthermore, the internal device **20** and the external device **50** may be connected by wires. In this configuration, for example, communication of an electrical signal converted into sound by the speaker **24** and supply of power can be performed by an electric wire. In this configuration, the communication unit **28** is unnecessary, and for example, the power source **26** is disposed in the external device **50**.

<Modification of Mounting Body>

As an example of the mounting body, a mounting body **122** illustrated in FIG. **3** may be used. The mounting body

122 includes a mounting portion 122A mounted on the upper dentition 92 of the wearer 90 and an attachment portion 122B arranged on the palate 94. The dotted portion in FIG. 3 corresponds to the attachment portion 122B.

The attachment portion 122B is a portion to which components such as the speaker 24 are attached. As a material of the attachment portion 122B, for example, a resin material such as polymethyl methacrylate, ethylene vinyl acetate (EVA), polyurethane, or polyethylene (PE) is used. The material of the attachment portion 122B is not limited to the above-described materials, and various materials can be used.

The mounting portion 122A is formed of a plurality of metal wires, and one end side thereof is fixed to the attachment portion 122B. The mounting portion 122A is attached by engaging the other end side of the mounting portion 122A with at least a part of the upper dentition 92 of the wearer 90. As a result, the attachment portion 122B is held in a state of being disposed on the palate 94. As the material of the metal wire constituting the mounting portion 122A, gold, silver, platinum, titanium, palladium, alloys thereof, or cobalt chromium and other wires are used.

Note that the mounting body 122 may not include the mounting portion 122A and may include only the attachment portion 122B. In this case, the attachment portion 122B is mounted in the oral cavity by its own adhesive force or the like, or is mechanically mounted and held in the oral cavity in a form of covering the dentition with the resin material.

Furthermore, when the wearer 90 uses a denture, the denture itself may be used as a mounting body.

<External Device 50>

The external device 50 is a device disposed outside the oral cavity of the wearer 90. The external device 50 is also a device operated by the wearer 90. The external device 50 is an example of a "device having a recording function". Specifically, the external device 50 is configured as a smartphone. Note that the external device 50 may be a tablet terminal or the like. Furthermore, the external device 50 may be a dedicated device, and various devices can be used. The external device 50 may be configured as a device separate from the vocalization assistance device 10.

As illustrated in FIG. 1, the external device 50 includes a housing 52 and an operation panel 54. Furthermore, as illustrated in FIG. 4, the external device 50 includes a microphone 56 and a control unit 60.

<Housing 52>

As illustrated in FIGS. 1 and 4, the housing 52 is a box (casing) that accommodates components such as the control unit 60. The housing 52 is formed in a panel shape (that is, a plate shape).

<Operation Panel 54>

The operation panel 54 is an operation unit to be operated by the wearer 90. The operation panel 54 is provided on one surface of the housing 52. Specifically, the operation panel 54 includes a touch panel display. The operation panel 54 can perform at least an output instruction to output sound from the speaker 24 and a stop instruction to stop the output of sound from the speaker 24.

<Microphone 56>

The microphone 56 is a device that converts sound into an electrical signal. The microphone 56 is used to record the sound output from the speaker 24. The sound recorded by the microphone 56 is converted into an electrical signal, and information of the electrical signal is recorded in a storage 63 described later of the control unit 60. Note that, when recording the voice of the wearer 90, it is desirable that the microphone 56 is disposed at a position where the speaker 24 (specifically, the outlet 24D) is disposed. As a result, the reproducibility of the voice can be enhanced. The voice of the wearer 90 to be recorded is, for example, the sound of "A".

<Control Unit 60>

The control unit 60 is a unit (that is, a control unit) that controls each unit of the external device 50 and the speaker 24. As illustrated in FIG. 4, the control unit 60 includes a computer including a processor 61, a memory 62, a storage 63, and a communication interface 64.

The communication interface 64 is an interface (that is, a communication unit) for communicating with the internal device 20. In the present embodiment, the communication interface 64 transmits an electrical signal converted into a sound by the speaker 24 to the communication unit 28 by a wireless communication means. As an example of the wireless communication means, a wireless communication function such as Bluetooth (registered trademark) is used.

Specifically, the processor 61 includes a central processing unit (CPU) that is a central processing unit. The storage 63 stores various programs including a control program 63A (see FIG. 5) and various pieces of data. Specifically, the storage 63 is realized by a recording device such as a hard disk drive (HDD), a solid state drive (SSD), and a flash memory.

The memory 62 is a work area for the processor 61 to execute various programs, and temporarily records various programs or various pieces of data when the processor 61 executes processing. The processor 61 reads various programs including the control program 63A from the storage 63 to the memory 62, and executes the program using the memory 62 as a work area.

In the control unit 60, the processor 61 executes the control program 63A to implement various functions. Hereinafter, a functional configuration realized by cooperation of the processor 61 as a hardware resource and the control program 63A as a software resource will be described. FIG. 5 is a block diagram illustrating a functional configuration of the processor 61, and is a block diagram mainly illustrating a functional configuration that realizes control of the operation panel 54.

As illustrated in FIG. 5, in the control unit 60, the processor 61 functions as an acquisition unit 61A and an output control unit 61B by executing the control program 63A.

The acquisition unit 61A acquires an instruction input by operating the operation panel 54. The instruction includes at least an output instruction to output sound from the speaker 24 and a stop instruction to stop output of sound from the speaker 24.

When the acquisition unit 61A acquires an output instruction to output sound from the speaker 24, the output control unit 61B causes the sound recorded by the microphone 56 to be output to the speaker 24. Specifically, when the acquisition unit 61A acquires the output instruction, the output control unit 61B reads the information of the electrical signal of the sound recorded by the microphone 56 from the storage 63, and transmits the electrical signal to the communication unit 28 of the internal device 20 through the communication interface 64. The speaker 24 converts the electrical signal received by the communication unit 28 into a sound and outputs the sound.

When the acquisition unit 61A acquires the stop instruction to stop the output of the sound of the speaker 24, the output control unit 61B stops the output of the sound from the speaker 24. Specifically, when the acquisition unit 61A acquires the stop instruction, the output control unit 61B stops transmitting the electrical signal to the communication unit **28** of the internal device **20** through the communication interface **64**.

Note that, by controlling the frequency of the electrical signal of the recorded sound, the speaker **24** may be configured to output sound subjected to processing such as intonation or height difference. For example, when an instruction operation for outputting an intonated sound from the speaker **24** is performed on the operation panel **54** and the instruction is acquired by the acquisition unit **61**A, the output control unit **61**B causes the speaker **24** to output the intonated sound. Note that, as the intonation, for example, an intonation in the case of speaking a question sentence or the like can be considered.

<Effects of Vocalization Assistance Device **10**>

Next, effects of the vocalization assistance device **10** will be described.

In the configuration of the vocalization assistance device **10**, the mouthpiece **22** is mounted in the oral cavity of the wearer **90**. The speaker **24** attached to the mouthpiece **22** is disposed along the person's palate **94** while the mouthpiece **22** is mounted in the oral cavity. Then, the speaker **24** can output a sound recorded by the external device **50** having a recording function as an original sound. As a result, the original sound output from the speaker is articulated into a verbal sound (a consonant and a vowel) by an articulatory organ such as a lip and a tongue, whereby utterance is performed. As described above, in the vocalization assistance device **10**, the internal device **20** is disposed in the oral cavity and thus cannot be seen from the surroundings. Therefore, the appearance is not impaired.

Here, since the speaker **24** can output the sound recorded by the external device **50** as the original sound, the degree of freedom in selecting the sound that can be used as the original sound is increased. For example, the voice of the wearer **90** can be used as the original sound by recording the voice of the wearer **90** with the external device **50**. Note that, in a case where the wearer **90** is a person who extracts the larynx including vocal cords by surgery or the like, the voice of the wearer **90** can be used as the original sound by recording his/her own voice before surgery. Note that the vocalization assistance device **10** may have a configuration in which an artificial sound recorded in advance on a chip or the like can be selectively used as the original sound in addition to the recorded sound.

In the present embodiment, the passage **24**C transmits the original sound output from the diaphragm **24**B to the rear side of the oral cavity. As a result, the original sound can be produced on the rear side of the oral cavity. As a result, in the case of making an explosive sound produced by releasing the tongue brought into contact with the palate, the original sound is easily blocked, and the explosive sound is easily made.

In the present embodiment, the passage **24**C emits the original sound transmitted to the rear side of the oral cavity from the outlet **24**D formed at the rear end of the case **24**A. As a result, the original sound can be produced on the rear side of the oral cavity. As a result, in the case of making an explosive sound produced by releasing the tongue brought into contact with the palate, the original sound is easily blocked, and the explosive sound is easily made.

In the present embodiment, the outlet **24**D through which the original sound is output is disposed at a position on the rear side of the rear end **94**A**1** of the hard palate **94**A of the wearer **90**.

Here, in a configuration in which the outlet **24**D of the speaker **24** is disposed at a position on the front side of the rear end **94A1** of the hard palate **94**A of the wearer **90** (hereinafter, referred to as a "first configuration"), for example, in a case of making an explosive sound produced by releasing the tongue brought into contact with the rear end **94A1** of the hard palate **94**A, the original sound is produced on the front side of a position at which the tongue comes into contact with and separates from the rear end **94A1** of the hard palate **94**A, and the original sound cannot be blocked. Therefore, it is difficult to make the explosive sound.

On the other hand, in the present embodiment, as described above, since the outlet **24**D is disposed at the position on the rear side from the rear end **94A1** of the hard palate **94**A of the wearer **90**, when the explosive sound is made, the original sound is produced on the rear side from the position where the tongue comes into contact with and separates from the rear end **94A1** of the hard palate **94**A, and thus the explosive sound can be articulated by the tongue. Therefore, as compared with the first configuration, it is possible to satisfactorily make the explosive sound.

Furthermore, in the present embodiment, the outlet **24**D is arranged at a position on the rear side from the contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of the consonant [k].

Here, in the configuration in which the outlet **24**D is disposed on the front side of the contact position where the tongue comes into contact with the palate **94** when the wearer makes the sound of consonant [k] (hereinafter, referred to as a "second configuration"), in a case where the wearer **90** makes a consonant [k], the original sound is produced on the front side of the position where the tongue comes into contact with and separates from the palate **94**, and the original sound cannot be blocked, so that it is difficult to make the consonant.

On the other hand, in the present embodiment, as described above, the outlet **24**D of the speaker **24** is disposed at the position on the rear side from the contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of consonant [k] in the mounting state of the mouthpiece **22**. Therefore, when the wearer makes the consonant [k], the original sound is produced on the rear side from the position where the tongue comes into contact with and separates from the palate **94**, and thus the explosive sound can be articulated by the tongue. Therefore, the consonant [k] can be made more satisfactorily than in the second configuration.

In the present embodiment, the outlet **24**D is disposed at a position on the rear side of the rearmost tooth **92**A of the wearer **90**.

Here, in the configuration in which the outlet **24**D is disposed at the position on the front side of the rearmost tooth **92**A of the wearer **90** (hereinafter referred to as "third configuration"), for example, in the case of making an explosive sound produced by releasing the tongue brought into contact with the palate **94** adjacent to the rearmost tooth **92**A, the original sound is produced on the front side of the position where the tongue comes into contact with and separates from the palate **94**, and the original sound cannot be blocked, so that it is difficult to make the explosive sound.

On the other hand, in the present embodiment, as described above, since the outlet **24**D is disposed at the position on the rear side from the rearmost tooth **92**A of the wearer **90**, when the explosive sound is made, the original sound is produced at the rear side, and thus, it is possible to articulate the explosive sound with the tongue. Therefore, the consonant can be made more satisfactorily than in the third configuration.

Further, in the present embodiment, the outlet **24**D is disposed at a position on the rear side from the rear end of the attachment portion **22**B. Therefore, the outlet **24**D can be disposed on the more rear side in the oral cavity.

In the present embodiment, the case **24**A of the speaker **24** protrudes rearward with respect to the attachment portion **22**B. Therefore, the outlet **24**D can be disposed on the more rear side in the oral cavity.

Here, in a case where the outlet **24**D is disposed on the more rear side in the oral cavity, there may be a new problem that the protruding portion of the case **24**A in which the outlet **24**D is formed comes into contact with the palate **94**, and the pharyngeal reflex is likely to occur. On the other hand, in the present embodiment, since the case **24**A including the protruding portion is not in contact with the palate **94**, the pharyngeal reflex hardly occurs.

Second Embodiment

A vocalization assistance device **200** according to a second embodiment will be described. FIGS. **6** and **7** are schematic views of the vocalization assistance device **200** according to the present embodiment.

The vocalization assistance device **200** is a device in which a configuration and an arrangement position of the speaker **24** and a configuration of the attachment portion **22**B of the mouthpiece **22** are changed from those of the vocalization assistance device **10** of the first embodiment. It should be noted that portions configured in the same manner as in the vocalization assistance device **10** of the first embodiment are denoted by the same reference numerals, and description thereof will be omitted as appropriate.

As illustrated in FIG. **6**, the vocalization assistance device **200** includes an internal device **220** and an external device **50**. The internal device **220** includes a mouthpiece **222**, the speaker **224**, a power source **26**, and a communication unit **28**.

<Mouthpiece **222**>

The mouthpiece **222** includes a mounting portion **22**A and an attachment portion **222**B facing the palate **94**. The attachment portion **222**B is an example of the facing portion. The mouthpiece **222** is configured similarly to the mouthpiece **22** except that the attachment portion **222**B is different in shape from the attachment portion **22**B of the mouthpiece **22** of the first embodiment. Note that a specific configuration of the attachment portion **222**B will be described later.

<Speaker **224**>

The speaker **224** is a device that converts an electrical signal (that is, electrical vibration) into a sound (that is, physical vibration). As illustrated in FIG. **7**, the speaker **224** includes a case **224**A, a diaphragm **224**B, and an amplifier (not illustrated).

As illustrated in FIGS. **6** and **7**, the case **224**A is formed in a flat shape that is thin in the vertical direction and expands in the horizontal direction and the front-rear direction. That is, the case **224**A is formed in a plate shape whose thickness direction is the vertical direction. As illustrated in FIG. **7**, an upper side (that is, a palate **94** side) of the case **224**A is open.

The diaphragm **224**B is accommodated in the case **224**A in a state of being exposed to the palate **94** side. The diaphragm **224**B can convert an electrical signal into a sound and output the original sound to the upper side (that is, the palate **94** side) through the opening of the case **224**A.

In the speaker **224**, after the electrical signal received by the communication unit **28** is amplified by the amplifier, the diaphragm **224**B converts the electrical signal into the original sound, and the original sound is output to the upper side (that is, the palate **94** side) through the opening of the case **224**A.

The size of the speaker **224** (specifically, the case **224**A) in a bottom view is at least smaller than the size of the attachment portion **222**B in a bottom view. Specifically, the speaker **224** (specifically, the case **224**A) has at least a width in the left-right direction smaller than a width in the left-right direction of the attachment portion **222**B. In other words, the speaker **224** (specifically, the case **224**A) has at least a width in the left-right direction narrower than an interval in the left-right direction between the left and right rearmost teeth **92**A of the wearer **90**. Note that the bottom view of the speaker **224** here is a bottom view when the attachment portion **222**B is seen through. The rearmost tooth **92**A is a tooth disposed on the rearmost side in the dentition **92**. Specifically, in the example of the present embodiment, the rearmost tooth **92**A is a second molar tooth. In the present embodiment, the length of the speaker **224** in the front-rear direction is shorter than the length of the attachment portion **222**B in the front-rear direction.

<Position of Speaker **224** and Specific Configuration of Attachment Portion **222**B>

The attachment portion **222**B has an attachment surface **23**A to which the speaker **224** is attached, and a projecting portion **23**B projecting upward (that is, toward the palate **94**). The speaker **224** is mounted to the surface of the attachment surface **23**A on the palate **94** side (that is, the upper surface).

The projecting portion **23**B is surrounded by the periphery (specifically, left side, right side, front side, and rear side) of the speaker **224**. The projecting portion **23**B projects to a position higher than the speaker **224** (that is, the position on the palate **94** side). Note that a hard resin material such as a denture resin may be used for the projecting portion **23**B. That is, the mouthpiece **222** may be made by changing a material for each part.

The attachment portion **222**B is further provided with a passage **21** that transmits the original sound output from the speaker **224** to the palate **94** side to the rear side. The passage **21** is formed by a space between the attachment portion **222**B and the palate **94**. Specifically, the passage **21** is formed by a space in a recess (that is, in a groove) provided on the upper surface side of the attachment portion **222**B from the attachment surface **23**A toward the rear. Therefore, the lower side and the left and right sides of the passage **21** are surrounded by the attachment portion **222**B, and the upper side is surrounded by the palate **94**. That is, the attachment portion **222**B has a lower wall, a left wall, and aright wall forming the passage **21**C. The space in the recess forming the passage **21** need not be a space completely closed by the palate **94**. The bottom surface of the recess forming the passage **21** is disposed at a position lower than the top of the projecting portion **23**B. In FIG. **7**, a transmission path of the original sound from the diaphragm **224**B is indicated by arrow **29**.

The passage **21** has an outlet **21**A through which the original sound transmitted to the rear side in the oral cavity of the wearer **90** is output. The outlet **21**A is disposed on the rear side in the oral cavity of the wearer **90** in the mounting state of the mouthpiece **222** (see FIGS. **6** and **7**).

In the present embodiment, the outlet **21**A is disposed at a position on the rear side of the rear end **94**A**1** of the hard palate **94**A of the wearer **90**. The position on the rear side of the rear end **94**A**1** of the hard palate **94**A is a position including the rear end **94**A**1** of the hard palate **94**A and a position on the rear side of the rear end **94**A**1** of the hard palate **94**A (that is, a soft palate **94**B present on the rear side of the hard palate **94**A). In the present embodiment, specifically, the outlet **21**A is disposed at a position on the rear side of the rear end **94A1** of the hard palate **94**A.

More specifically, the outlet **21**A is disposed at a position on the rear side from the rearmost tooth **92**A of the wearer **90**. The position on the rear side from the rearmost tooth **92**A is a position including the rearmost tooth **92**A and the rear side of the rearmost tooth **92**A. For the wearer **90** having no second molar tooth, the outlet **21**A is arranged at a position on the rear side from the position where a second molar tooth is arranged in a case where the second molar tooth is present.

Furthermore, the outlet **21**A is disposed at a position on the rear side from a contact position at which the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of consonant [k]. The position on the rear side from the contact position is a position including the contact position and a position on the rear side of the contact position. Furthermore, in the present embodiment, the outlet **21**A is disposed at the rear end of the attachment portion **222**B.

Note that the position of the hard palate **94**A, the position of the rearmost tooth **92**A, the position where the tongue comes into contact with the palate **94** when making the sound of the consonant [k], and the like have individual differences, and thus, for example, the position of the outlet **21**A and the like are adjusted in accordance with each wearer **90**.

A plurality of speakers **224** may be provided at the attachment portion **222**B. As a result, the range of sound that can be made can be expanded. Furthermore, the speaker **224** may include a plurality of diaphragms **224**B. As a result, the range of sound that can be made can be expanded.

In the second embodiment, similarly to the mounting body **122** (see FIG. **3**) of the first embodiment, a mounting body in which the attachment portion is formed of a plurality of metal wires and a mounting body which does not have the attachment portion and is formed of only the attachment portion may be used.

<Effects of Vocalization Assistance Device **200**>

Next, effects of the vocalization assistance device **200** will be described.

In the vocalization assistance device **200**, the same effects as those of the vocalization assistance device **10** are obtained. Specifically, in the vocalization assistance device **200**, the passage **21** transmits the original sound output from the diaphragm **224**B to the rear side of the oral cavity. As a result, the original sound can be produced on the rear side of the oral cavity. As a result, in the case of making an explosive sound produced by releasing the tongue brought into contact with the palate, the original sound is easily blocked, and the explosive sound is easily made.

Furthermore, in the present embodiment, as described above, since the outlet **21**A is arranged at the position on the rear side from the rear end **94A1** of the hard palate **94**A of the wearer **90**, in a case where the explosive sound is made, the original sound is produced on the rear side from the position where the tongue comes into contact with and separates from the rear end **94A1** of the hard palate **94**A, and thus, it is possible to articulate the explosive sound with the tongue. Therefore, as compared with the configuration in which the outlet **21**A is disposed at the position on the front side of the rear end **94A1** of the hard palate **94**A of the wearer **90**, the explosive sound can be satisfactorily made.

Furthermore, in the present embodiment, since the outlet **21**A is arranged at the position on the rear side from the contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of the consonant [k], in a case where the wearer makes the consonant [k], the original sound is produced on the rear side from the position where the tongue comes into contact with and separates from the palate **94**, and thus the explosive sound can be articulated by the tongue. Therefore, as compared with the configuration in which the outlet **21**A is disposed on the front side of the contact position where the tongue comes into contact with the palate **94** when the wearer **90** makes the sound of consonant [k], it is possible to satisfactorily make the consonant [k].

Furthermore, in the present embodiment, as described above, since the outlet **21**A is arranged at the position on the rear side from the rearmost tooth **92**A of the wearer **90**, when the explosive sound is made, the original sound is produced at the more rear side, and thus, it is possible to articulate the explosive sound with the tongue. Therefore, as compared with the configuration in which the outlet **21**A is disposed at the position on the front side of the rearmost tooth **92**A of the wearer **90**, the consonant can be satisfactorily made.

In the present embodiment, the outlet **21**A is disposed up to the rear end of the attachment portion **22**B. Therefore, the outlet **21**A can be disposed on the more rear side in the oral cavity.

Further, in the configuration of the vocalization assistance device **200**, as described above, the speaker **224** is attached to the surface (that is, the upper surface) of the attachment surface **23**A of the attachment portion **222**B on the side of the palate **94**. As a result, the lower side of the speaker **224** is covered with the attachment portion **222**B, and water such as saliva accumulated in the oral cavity and the speaker **224** are less likely to touch, so that water is less likely to enter the speaker **224**.

In addition, even in a case where water such as saliva enters the case **224**A of the speaker **224**, the vocalization assistance device **200** can discharge the water through the opening at the upper end of the case **224**A.

Third Embodiment

A vocalization assistance device **300** according to a third embodiment will be described. FIGS. **8**, **9**, and **10** are schematic views of the vocalization assistance device **300** according to the present embodiment.

The vocalization assistance device **300** is a device in which, in the vocalization assistance device **10** according to the first embodiment, the attachment portion **322**B to which the speaker **24** is attached is vertically movable. It should be noted that portions configured in the same manner as in the vocalization assistance device **10** of the first embodiment are denoted by the same reference numerals, and description thereof will be omitted as appropriate.

As illustrated in FIG. **8**, the vocalization assistance device **300** includes an internal device **320** and an external device **50**. The internal device **320** includes a mouthpiece **322**, a speaker **24**, a power source **26**, and a communication unit **28**.

<Mouthpiece **322**>

The mouthpiece **322** includes a mounting portion **322**A and an attachment portion **322**B facing the palate **94**. The attachment portion **322**B is an example of the facing portion. The mouthpiece **322** is configured similarly to the mouthpiece **22**, except that the mounting portion **322**A and the attachment portion **322**B are different in shape from the mounting portion **22**A and the attachment portion **22**B of the mouthpiece **22** of the first embodiment.

In the mounting portion **322**A, a portion on the front side is notched. Therefore, the mounting portion **322**A is configured not to be attached to some teeth on the front side but to be attached to some teeth on the rear side in the dentition **92**. The attachment portion **322**B does not cover the front portion of the palate **94**, so that the front portion of the palate **94** is exposed downward.

As described above, in the present embodiment, the front portions of the mounting portion **322**A and the attachment portion **322**B are notched, and a part of the teeth on the front side of the dentition **92** and the front portion of the palate **94** are exposed.

Further, the attachment portion **322**B is connected to the mounting portion **322**A at a front side portion, and is separated from the mounting portion **322**A at a rear side portion. As a result, as illustrated in FIG. **9**, the attachment portion **322**B is movable in the vertical direction with the front side as a fulcrum.

<Effects of Vocalization Assistance Device **300**>

Next, effects of the vocalization assistance device **300** will be described.

The vocalization assistance device **300** also has the following effects in addition to the effects of the vocalization assistance device **10**. In the configuration of the vocalization assistance device **300**, the attachment portion **322**B is movable in the vertical direction with the front side as a fulcrum. Therefore, the direction of the outlet **24**D of the speaker **24** can be changed, and the direction in which the original sound is output from the speaker **24** can be changed.

In the present embodiment, for example, due to the weight of the speaker **24**, the speaker **24** is positioned at an intermediate position between the tongue and the palate **94** (a position indicated by a solid line in FIG. **9**). As a result, the original sound can be transmitted further to the rear side (that is, the back side) in the oral cavity, and it is possible to vocalize a nasal sound and the like.

The output direction from the outlet **24**D of the speaker **24** is, for example, a direction indicated by arrow **329**. Further, an opening may be formed on the side of the palate **94** (that is, the upper side) in a portion on the rear side in the case **24**A of the speaker **24**, so that the original sound is output toward the palate **94** (see arrow **333**). In this case, for example, by making the size of the opening smaller than the size of the opening at the rear end of the case **24**A, the loudness of the original sound output to the upper side (see arrow **333**) can be made smaller than the loudness of the original sound output to the rear side (see arrow **329**).

When it is necessary to bring the tongue into contact with the palate **94**, the tongue can be brought into contact with the palate **94** while moving the speaker **24** upward with the tongue. As a case where it is necessary to bring the tongue into contact with the palate **94**, there is a case where an explosive sound is produced by releasing the tongue brought into contact with the palate **94** from the palate **94**.

In the present embodiment, as described above, the front portions of the mounting portion **322**A and the attachment portion **322**B are notched, and a part of the teeth on the front side of the dentition **92** and the front portion of the palate **94** are exposed.

Therefore, contact between a part of the teeth on the front side and the front portion of the palate **94** and the tongue is not prevented, so that frictional sounds such as consonant [s] are easily made. The configuration in which the attachment portion and the front portion of the attachment portion are notched may be applied to the first embodiment and the second embodiment.

<Modification of Speaker **24**>

In the present embodiment, as illustrated in FIG. **10**, the above-described speaker **24** that outputs the original sound upward may be used instead of the speaker **224**. In the present modification, the direction in which the original sound is output from the speaker **224** can be changed.

In the present embodiment, for example, due to the weight of the speaker **224**, the speaker **224** is positioned at an intermediate position between the tongue and the palate **94** (a position indicated by a solid line in FIG. **10**). As a result, the original sound can be transmitted further to the rear side (that is, the back side) in the oral cavity, and it is possible to vocalize a nasal sound and the like.

The direction in which the original sound is output from the speaker **224** is, for example, a direction indicated by arrow **335**. Further, the original sound may be output to the rear side by forming a rear side opening in the case **224**A of the speaker **224** (see arrow **337**). In this case, for example, by making the size of the opening smaller than the size of the opening on the upper side of the case **24**A, the loudness of the original sound output to the rear side (see arrow **337**) can be made smaller than the loudness of the original sound output to the upper side (see arrow **335**).

When it is necessary to bring the tongue into contact with the palate **94**, the tongue can be brought into contact with the palate **94** while moving the speaker **224** upward with the tongue. As a case where it is necessary to bring the tongue into contact with the palate **94**, there is a case where an explosive sound is produced by releasing the tongue brought into contact with the palate **94** from the palate **94**.

A passage **31** is formed by a space between the case **224**A and the palate **94** and a space between the attachment portion **322**B and the palate **94** on the rear side with respect to the diaphragm **224**B in a state where the speaker **224** moves upward.

<Modification of Mouthpiece **322**>

An example of the mounting body is not limited to the mouthpiece **322**. For example, in the mouthpiece **322**, a part on an outer peripheral side (that is, a buccal side) may be a mounting body made of a hard resin material such as a denture resin. In the mouthpiece **322**, the mounting portion **322**A may be a mounting body using a hard resin material such as a denture resin.

As the mounting body, a mounting body to which the attachment portion **322**B (that is, a movable portion) is attached so as to be movable in the vertical direction with the front side as a fulcrum with respect to a support member (so-called palatal bar) stretched between the left tooth and the right tooth in the dentition **92** along the palate **94** may be used.

Furthermore, when the wearer **90** uses a denture, a mounting body in which the attachment portion 322B (that is, a movable portion) is attached to the denture so as to be movable in the vertical direction with the front side as a fulcrum may be used.

Although one embodiment of the present invention has been described above, the present invention is not limited to the above, and it is needless to say that various modifications can be made in addition to the above without departing from the gist of the present invention.

For example, a plurality of the above-described embodiments and modifications may be appropriately combined.

<Supplementary Note>

A first aspect provides a vocalization assistance device including: a mounting body that is mounted in an oral cavity of a person and that has a facing portion facing a palate of the person; a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound; and a passage that is provided at the facing portion and that is configured to transmit the original sound output from the diaphragm to a rear side of the oral cavity.

A second aspect is the vocalization assistance device according to the first aspect, in which the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and the outlet is provided at a position at a rear side from a rear end of a hard palate of the person.

A third aspect is the vocalization assistance device according to the first aspect, in which the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and the outlet is provided at a position at a rear side from a position at which a tongue comes into contact with the palate in a case in which the person makes a consonant [k].

A fourth aspect is the vocalization assistance device according to the first aspect, in which the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and the outlet is provided at a position at a rear side from a rearmost tooth of the person.

A fifth aspect is the vocalization assistance device according to the first aspect, in which the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and the outlet is provided at a position at a rear side from a rear end of the facing portion.

A sixth aspect is the vocalization assistance device according to any one of the first to fifth aspects, in which the diaphragm is disposed on a surface of the facing portion on a palate side, and is capable of outputting the original sound to the palate side.

A seventh aspect is the vocalization assistance device according to the sixth aspect, in which the passage is formed by a space between the facing portion and the palate of the person.

An eighth aspect is the vocalization assistance device according to the sixth or seventh aspect, further including: a case in which the palate side is open and in which the diaphragm is accommodated in a state of being exposed to the palate side.

A ninth aspect is the vocalization assistance device according to any one of the first to fifth aspects, further including: a case in which the diaphragm is accommodated and in which the passage is formed, wherein: the passage emits the original sound transmitted to the rear side of the oral cavity from an outlet formed at a rear end of the case.

A tenth aspect is the vocalization assistance device according to the ninth aspect, in which the case protrudes rearward with respect to the facing portion.

An eleventh aspect is the vocalization assistance device according to the tenth aspect, in which a rearward protruding portion of the case is not in contact with the palate.

A twelfth aspect is the vocalization assistance device according to any one of the first to eleventh aspects, in which the mounting body is a mouthpiece.

A thirteenth aspect is the vocalization assistance device according to any one of the first to twelfth aspects, in which the facing portion is movable in a vertical direction with a front side of the oral cavity as a fulcrum.

A fourteenth aspect provides a vocalization assistance device including: a mounting body that is mounted in an oral cavity of a person, that has a facing portion facing a palate of the person, and that is movable in a vertical direction with a front side as a fulcrum; and a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound.

The disclosure of Japanese Patent Application No. 2021-010291 filed on Jan. 26, 2021 is incorporated herein by reference in its entirety. All documents, patent applications, and technical standards described in this specification are incorporated herein by reference to the same extent as if each individual document, patent application, and technical standard were specifically and individually indicated to be incorporated by reference.

The invention claimed is:

1. A vocalization assistance device comprising:
- a mounting body that is mounted in an oral cavity of a person and that has a facing portion facing a palate of the person;
- a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound; and
- a passage that is provided at the facing portion and that is configured to transmit the original sound output from the diaphragm to a rear side of the oral cavity.

2. The vocalization assistance device according to claim **1**, wherein:
- the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and
- the outlet is provided at a position at a rear side from a rear end of a hard palate of the person.

3. The vocalization assistance device according to claim **1**, wherein:
- the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and
- the outlet is provided at a position at a rear side from a position at which a tongue comes into contact with the palate in a case in which the person makes a consonant [k].

4. The vocalization assistance device according to claim **1**, wherein:
- the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and
- the outlet is provided at a position at a rear side from a rearmost tooth of the person.

5. The vocalization assistance device according to claim **1**, wherein:
- the passage has an outlet through which the original sound transmitted to the rear side of the oral cavity is emitted, and
- the outlet is provided at a position at a rear side from a rear end of the facing portion.

6. The vocalization assistance device according to claim **1**, wherein the diaphragm is disposed on a surface of the facing portion on a palate side, and is capable of outputting the original sound to the palate side.

7. The vocalization assistance device according to claim **6**,
- wherein the passage is formed by a space between the facing portion and the palate of the person.

8. The vocalization assistance device according to claim **6**, further comprising:
- a case in which the palate side is open and in which the diaphragm is accommodated in a state of being exposed to the palate side.

9. The vocalization assistance device according to claim **1**, further comprising:

a case in which the diaphragm is accommodated and in which the passage is formed, wherein:

the passage emits the original sound transmitted to the rear side of the oral cavity from an outlet formed at a rear end of the case.

10. The vocalization assistance device according to claim **9**, wherein the case protrudes rearward with respect to the facing portion.

11. The vocalization assistance device according to claim **10**, wherein a rearward protruding portion of the case is not in contact with the palate.

12. The vocalization assistance device according to claim **1**, wherein the mounting body is a mouthpiece.

13. The vocalization assistance device according to claim **1**, wherein the facing portion is movable in a vertical direction with a front side of the oral cavity as a fulcrum.

14. A vocalization assistance device comprising:

a mounting body that is mounted in an oral cavity of a person, that has a facing portion facing a palate of the person, and that is movable in a vertical direction with a front side as a fulcrum; and

a diaphragm that is provided at the facing portion and that is capable of outputting a sound recorded by a device having a recording function, as an original sound.

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